

EXPERIMENT – PICK AND PLACE OBJECTS USING IRB360 IN MANUAL MODE

AIM

To program IRB360 flex picker 4 axis parallel manipulator using manual mode

EQUIPMENT REQUIRED

IRB 360 Flex picker
Conveyer assembly with VFD

PROCEDURE:

- Give the pre - electrical supplies to the robot.
- White light will start blinking.
- Grip the flex pendant appropriate and press the end
- Select jogging mode. Before doing the above, check if manual mode is selected.
- Axis, Linear modes need to be used appropriately along with parameters such as speed is adjusted as per need.
- The gripper must be opened before the exact position of gripping.
- Go to ABB main menu and go to >Input and Output.
- Set do_vaccum1 and reset do_vaccum1 to release the gripper.
- The gripper grips the coins and slowly lifts its arm.
- To open the grippers go to Input and Output and reset do_vaccum1.
- Move the arm to a safe position after the object is dropped at the drop point.



Figure-01 4 axis parallel manipulator (flex picker)

Safety precautions:

- Careful in operating the robot as the end effector is sensitive rubber latex gripper avoid collision with the conveyor belt
- If it happens use relief button to replace each link in safer position

Prerequisites :

- While operating the robot one should go through with the safety and instruction/operation manual of the robot in order to follow safety.

RESULT

Thus, to program IRB360 flex picker 4 axis parallel manipulator using manual mode is successfully completed.